

XCS224R Assignment 1

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XCSR224R

1.b Behavior Cloning

Task Name	Eval Mean	Eval Standard Deviation
Ant-v4	3827.21	1613.09
Hopper-v4	889.08	63.05

```

Saving rollouts as videos...
Eval_AverageReturn : 3827.205810546875
Eval_StdReturn : 1613.0872802734375
Eval_MaxReturn : 4701.90625
Eval_MinReturn : 31.736379623413086
Eval_AverageEpLen : 844.25
Train_AverageReturn : 4713.6533203125
Train_StdReturn : 12.196533203125
Train_MaxReturn : 4725.849609375
Train_MinReturn : 4701.45654296875
Train_AverageEpLen : 1000.0
Train_EnvStepsSoFar : 2000
TimeSinceStart : 15.197916269302368
Training Loss : -15.896620750427246
Initial_DataCollection_AverageReturn : 4713.6533203125
Done logging...

```

Figure 1: Ant-v4 results

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Saving rollouts as videos...
Eval_AverageReturn : 889.0819091796875
Eval_StdReturn : 63.054203033447266
Eval_MaxReturn : 1175.9554443359375
Eval_MinReturn : 844.8524169921875
Eval_AverageEpLen : 262.28205128205127
Train_AverageReturn : 3772.67041015625
Train_StdReturn : 1.9483642578125
Train_MaxReturn : 3774.61865234375
Train_MinReturn : 3770.721923828125
Train_AverageEpLen : 1000.0
Train_EnvStepsSoFar : 2000
TimeSinceStart : 11.965486764907837
Training Loss : -4.396175861358643
Initial_DataCollection_AverageReturn : 3772.67041015625
Done logging...

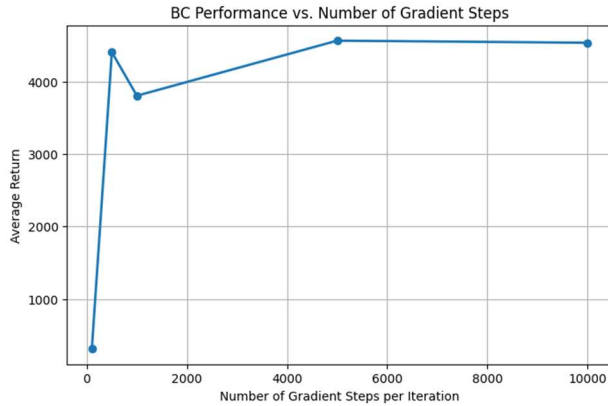
```

Figure 2: Hopper-v4 results

For this assignment Ant-v4 and Hopper-v4 were chosen to complete this assignment. Both environments were run with `n_iter=1`, `eval_batch_size=10,000` and `ep_len=1,000`. Behavioral Cloning (BC) performed very differently across the two MuJoCo environments. On Ant-v4, the cloned policy achieved an average return of approximately 3827, which is about 81% of the expert's performance. The agent was able to remain stable for most of the episode, demonstrating that BC can work well when the environment is relatively forgiving and the expert trajectories cover a wide range of states. In contrast, on Hopper-v4, it achieved an average return of only 889, roughly 23% of the expert's performance, below 30% with episodes terminated early (average length ≈ 262 steps) due to falling. This indicates that BC struggled to generalize in Hopper-v4, possibly due to the highly unstable single-legged system. These results show that BC's success is highly environment-dependent and very sensitive to the stability of the underlying system and dynamics (stable multi-legged Ant versus unstable single-legged Hopper).

1.c Behavior Cloning Experiment

For this task this paper chose *num_agent_train_steps_per_iter* as a hyperparameter. The number of gradient steps controls how much the BC policy fits the expert data, with too few steps, the policy might underfit, and with too many it may overfit or show no improvements. We did a sweep as follows [100, 500, 1000, 5000, 10000] with Figure 3 showing the results. Varying the number of gradient steps per iteration had a strong effect on Behavioral Cloning performance. With only 100 training steps, the policy severely underfit the expert data and achieved a return of



just ~312. Increasing the number of steps to 500 produced a dramatic improvement to ~4406, indicating that the model had finally learned a meaningful approximation of the expert policy. Beyond 1000 steps, performance showed diminishing improvement.

Figure 3: Diminishing returns > 1000 steps

2.b DAgger

To compare BC performance to DAgger on the Ant-v4 environment. DAgger was run for *n_iter=10*, *eval_batch_size=10,000* and *ep_len = 1,000*. We can see that DAgger started matching BC performing but after just one iteration it was able to improve and after three iteration it was able to match Expert performance. See Figure 4.

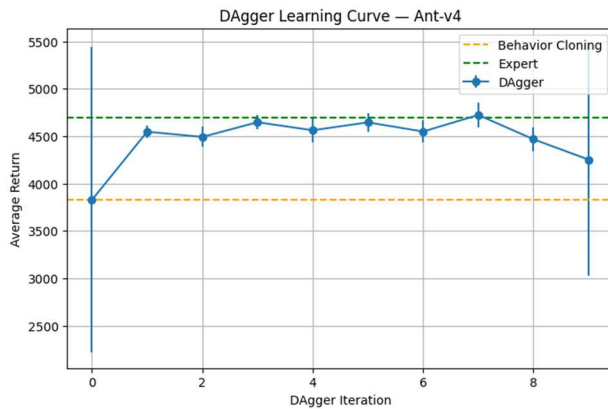


Figure 4: DAgger and Ant-v4

Deep Reinforcement Learning

Next we compare DAgger on the Hopper-v4 environment. Using the same settings: $n_iter=10$, $eval_batch_size=10,000$ and $ep_len = 1,000$. Again here we see a substantial improvement over BC, where DAgger was at times able to match Expert performance, although, the performance is less stable (see Figure 5). This is possibly due to the inherent instability of the system (one-legged).

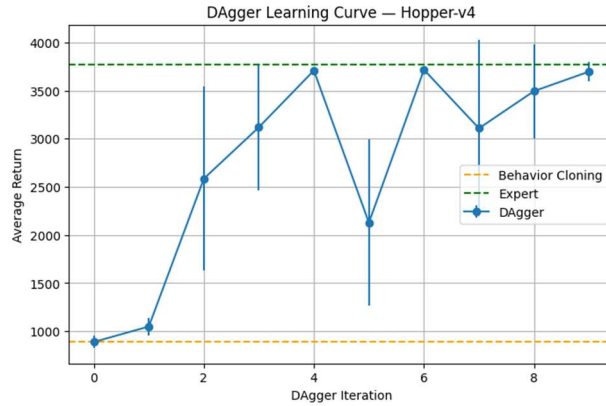


Figure 5: DAgger on Hopper-v4

The above experiments clearly show an improved performance of DAgger over BC. It proves the superiority of letting the agent interact with the world.